Harshith Reddy Madireddy

AI/ML Researcher · Fall 2027 MS/PhD Applicant

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Hyderabad, India

ACADEMIC PROFILE

AI/ML researcher investigating predictive cybersecurity, continuous dynamical modeling, and explainable machine learning architectures. Awarded the INEX 2024 International Silver Medal for a 99.255% accurate 2D-wavelet spatial intrusion detection framework, and author of continuous second-order differential equation early-warning modeling presented at IIT Dhanbad. Experienced in production AI infrastructure and enterprise regulatory automation—preparing for Fall 2027 MS/PhD study.

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad, India (Nov 2021 – May 2025)
Bachelor of Technology, Information Technology · **GPA: 8.22 / 10.0 (3.53 / 4.0)**

PUBLICATIONS & PREPRINTS

- **A dual-pillar framework for DDoS detection combining ODE-inspired early-warning signal analysis with wavelet-based machine learning classification**
  Harshith Reddy Madireddy.
  Research Square (Preprint), 2026. Forthcoming in major journals / conferences.
  Permanent DOI: doi.org/10.5281/zenodo.23671072

AWARDS, PRESENTATIONS & OLYMPIAD MEDALS

- **2nd Place · Silver Medal** — India International Innovation and Invention Expo (INEX 2024), Indian Innovators Association: Awarded 2nd place among 150+ international innovators from 15+ countries for 2D wavelet transform-based spatial network intrusion detection (99.255% accuracy).
- **SIPHO Physics Olympiad — National Rank 190 · Gold Medal:** Recognized for competitive problem solving in physical systems, analytical mechanics, and classical dynamics.
- **SIMO Mathematics Olympiad — National Rank 235 · Gold Medal:** Recognized for analytical problem solving in mathematical logic, algebra, and discrete structures.
- **Presented Research** — "ODE+ML=EWS: Early Warning Signals for Cyber Attacks" at National Conference on Data Predictive Analytics and Numerical Simulations (DPA-NS 2025), Indian Institute of Technology (ISM) Dhanbad.
- **Presented Research** — "Dynamical Modeling in Network Security & Early Warning" at Industrial Innovation and Technology Expo (IITEX 2025), The Federation of Telangana Chambers of Commerce and Industry (FTCCI).
- **Excellence in Community Service Award (2026):** Community Service & Outreach — Recognized for sustained community service, educational outreach, and local welfare initiatives.

INDUSTRY & RESEARCH EXPERIENCE

Pulse Platform — AI Engineer · Part-time → Full-time (August 2025 – Present, Seattle, USA · Remote)

- Contributed to the AI signal infrastructure powering real-time market intelligence across news, social and financial data sources.
- Worked on sentiment analysis, multi-agent consensus and dynamic risk scoring with 94% signal accuracy and up to 85% predictive reliability.
- Built low-latency streaming workflows for real-time signal generation and delivery for 200+ users in testing/early access.

Thermo Fisher Scientific — Software Intern · Regulatory Affairs R&D (April 2025 – March 2026, Hyderabad, India)

- Architected an Enterprise RAG Pipeline using LangChain and Azure OpenAI to support medical device registration across 40+ international jurisdictions, reducing manual overhead by ~90%.
- Strengthened medical device cybersecurity through systematic threat modeling, hazard analysis, and standard operating procedures aligned with regulatory standards.

VNR Vignana Jyothi Institute of Engineering and Technology — Undergraduate Student Researcher (April 2024 – April 2025, Hyderabad, India)

- Formulated a 2D-wavelet spatial intrusion detection system achieving 99.255% classification accuracy (INEX 2024 Silver Medal).
- Investigated a second-order dynamical system formulation to characterize rate-of-change curvature anomalies in packet arrivals ~83 minutes prior to peak flood on evaluated testing data.

IBM SkillsBuild — Web Developer Intern (June 2023 – July 2023)

- Developed responsive full-stack web applications using React.js, Node.js, JavaScript, HTML5/CSS3, and database management systems.
- Implemented modular user interfaces ensuring cross-platform responsiveness, managed code versioning with Git, and collaborated via agile workflows.

Cisco Networking Academy — Cyber Security Engineering Intern (April 2023 – May 2023)

- Completed practical training in digital security operations, threat modeling, and incident response in simulated network environments.
- Analyzed attack surfaces, network vulnerabilities, and defensive countermeasure workflows.

KEY FEATURED PROJECTS

- **Spatial 2D-Wavelet DDoS Detection System (99.255% Accuracy):** High-precision intrusion detection decomposing 58 network telemetry attributes into 8×8 spatial-frequency matrices via 2D Haar DWT and Extra Trees classifier.
  `Interactive Research System for Detection of DDoS:` detect-ddos.onrender.com
- **Second-Order ODE Early Warning System (~83 min Lead Time):** Continuous dynamical system harness modeling benign network rate-of-change dynamics to identify precursor velocity anomalies ~83 minutes prior to volumetric flood on evaluated testing data.
  `Interactive Research System for EWS:` ddos-ews.onrender.com
- **CypherDocs - Hybrid Graph-Vector RAG Platform:** Built enterprise-grade GraphRAG integrating Neo4j knowledge graphs and vector embeddings for relationship-aware semantic retrieval across unstructured documents (FastAPI, Neo4j, Gemini API, Docker).
- **Social Media Intelligence Hub:** Modular pipeline combining encoder models (DistilBERT) for sentiment classification and generative LLMs (GPT-2) for automated summarization (FastAPI, Hugging Face, PyTorch).

PROFESSIONAL CERTIFICATIONS & CREDENTIALS

- **Microsoft Azure AI Essentials Professional Certificate** — Microsoft (2025)
- **Data Science Professional Certificate** — KNIME (2025)
- **Supervised Machine Learning: Regression and Classification** — DeepLearning.AI (2024)
- **Docker Foundations Professional Certificate** — Docker, Inc. (2025)
- **Prompt Engineering and Programming with OpenAI** — Columbia+ (2025)
- **Public Speaking Skills Professional Certificate** — Toastmasters International (2025)
- **Agile Virtual Experience Program** — JPMorgan Chase & Co. (2023)

ATHLETICS & SPORTS CONTRIBUTIONS

- **University Badminton:** First Place · Gold Medal (Inter-university Doubles Champion).
- **Inter-School Kabaddi Competition:** First Place · Gold Medal (Team Captain).

VOLUNTEER EXPERIENCE & COMMUNITY LEADERSHIP

Hussaini Welfare Association — Education & Outreach Lead (Pro Bono, June 2020 – Present): Led sustained education and community initiatives across mentorship, healthcare, and environmental programs, mentoring 15 students over four consecutive years, coordinating health camps serving 200+ residents, and mobilizing 20+ volunteers for environmental initiatives.

TECHNICAL SKILLS

- **Core Languages:** Python, C++, SQL, JavaScript
- **ML & Application Frameworks:** PyTorch, Scikit-learn, Hugging Face, LangChain, FastAPI
- **Data & Systems Infrastructure:** Neo4j (Cypher), Docker, Azure OpenAI, Git / CI/CD
- **Mathematical & Analytical Methods:** Differential Equations, Dynamical Systems, Wavelet Transforms, Numerical Simulation